Calculus and Linear Algebra

Zeros, Intersections and Sign

Lecture 12

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Lecture outline

- Domain of composite functions.
- Zeros and intersections with the coordinate axes.
- Intersections between two functions.
- Sign of a function and sign charts.
- Polynomial, rational, radical, logarithmic and absolute-value examples.
- Guided and independent practice.

Main goal

After finding where a function exists, determine **where it is zero, positive or negative**, and where its graph meets other graphs.

Domain of a composite function

If

$$f : A \to B, \qquad g : B \to C,$$

then

$$(g \circ f)(x) = g(f(x)).$$

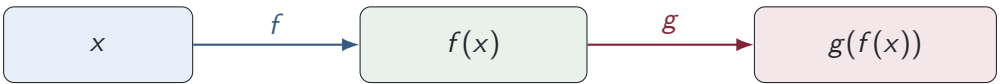
For the composition to be defined, two conditions are required:

$$\boxed{x \in D_f \qquad \text{and} \qquad f(x) \in D_g.}$$

Therefore

$$\boxed{D_{g \circ f} = \{x \in D_f : \ f(x) \in D_g\}.}$$

Composition: think in two steps



1. Can we insert x into f?
2. Is the output $f(x)$ an admissible input for g?

The second condition is the one most often forgotten.

The formula of $g \circ f$ may create a new restriction that was not visible in D_f alone.

Composite domain: square root after a polynomial

Let

$$f(x) = x^2 - 4, \qquad g(u) = \sqrt{u}.$$

Then

$$(g \circ f)(x) = \sqrt{x^2 - 4}.$$

Since g accepts only non-negative inputs,

$$f(x) = x^2 - 4 \geq 0.$$

Thus

$$(x-2)(x+2) \geq 0 \quad \Longrightarrow \quad \boxed{D_{g\circ f} = (-\infty, -2] \cup [2, +\infty).}$$

Composite domain: logarithm after a polynomial

Let

$$f(x) = 3 - x^2, \qquad g(u) = \ln u.$$

Then

$$(g \circ f)(x) = \ln(3 - x^2).$$

The logarithm requires

$$3 - x^2 > 0 \quad \Longleftrightarrow \quad x^2 < 3.$$

Hence

$$\boxed{D_{g \circ f} = (-\sqrt{3}, \sqrt{3}).}$$

Notice the strict inequality: the argument of a logarithm cannot be zero.

Nested functions: a useful example

Consider

$$h(x) = \sqrt{\ln x}.$$

Work from the outside inward. The square root requires

$$\ln x \geq 0.$$

This already implies $x > 0$, and

$$\ln x \geq 0 \quad \Longleftrightarrow \quad x \geq 1.$$

Therefore

$$\boxed{D_h = [1, +\infty).}$$

Method

For nested functions, translate the domain of the **outer** function into a condition on the output of the inner function.

Quick check: composite domains

Exercise Find the natural domain of

$$\text{A: } \sqrt{4-x^2}, \qquad \text{B: } \ln(x^2-1), \qquad \text{C: } \frac{1}{\sqrt{2-x}}.$$

Quick check: composite domains

Exercise Find the natural domain of

$$\text{A: } \sqrt{4-x^2}, \qquad \text{B: } \ln(x^2-1), \qquad \text{C: } \frac{1}{\sqrt{2-x}}.$$

Solution

$$D_A = [-2, 2],$$
$$D_B = (-\infty, -1) \cup (1, +\infty),$$
$$D_C = (-\infty, 2).$$

In C the radicand must be **strictly positive** because the root is in the denominator.

From domain to qualitative analysis

For every function, we now separate three questions:

1. Where does f exist?

2. Where is $f(x) = 0$?

3. Where is $f(x) > 0$ or $f(x) < 0$?

Do not mix them

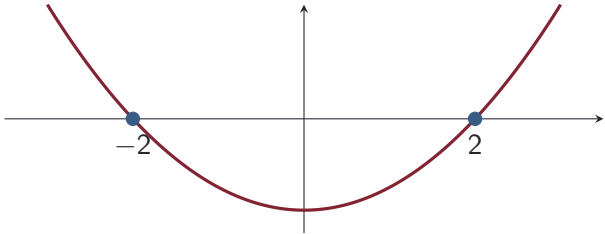
Domain restrictions, zeros and sign changes play different roles in a function study.

Zeros of a function

Definition A number $x_0 \in D_f$ is a **zero** (or root) of f if

$$f(x_0) = 0.$$

Geometrically, zeros are the x-coordinates of the points where the graph intersects the x-axis.



For $f(x) = x^2 - 4$, the zeros are $x = -2$ and $x = 2$.

Intersections with the coordinate axes

For a graph $y = f(x)$:

x-axis

Set

$$f(x) = 0.$$

Every solution x_i gives the point

$$(x_i, 0).$$

y-axis

Set

$$x = 0.$$

If $0 \in D_f$, the intersection is

$$(0, f(0)).$$

A function may have no x-intercepts, one, or several; it has at most one y-intercept.

Example: intercepts of a quadratic

Let

$$f(x) = x^2 - x - 6 = (x-3)(x+2).$$

x-intercepts:

$$(x-3)(x+2) = 0 \quad \Longrightarrow \quad x = 3 \text{ or } x = -2.$$

Thus

$$(-2, 0), \qquad (3, 0).$$

y-intercept:

$$f(0) = -6,$$

so

$$(0, -6).$$

Intersections between two functions

The graphs of $y = f(x)$ and $y = g(x)$ intersect when they have the same output for the same input:

$$\boxed{f(x) = g(x).}$$

After solving this equation, substitute each solution into either function to obtain the corresponding y-coordinate.

Domain first

A candidate intersection is valid only if the input belongs to the domains of **both** functions.

Example: a line and a parabola

Find the intersections of

$$f(x) = x^2, \qquad g(x) = x + 2.$$

Solve

$$x^2 = x + 2 \quad \Longleftrightarrow \quad x^2 - x - 2 = 0 \quad \Longleftrightarrow \quad (x-2)(x+1) = 0.$$

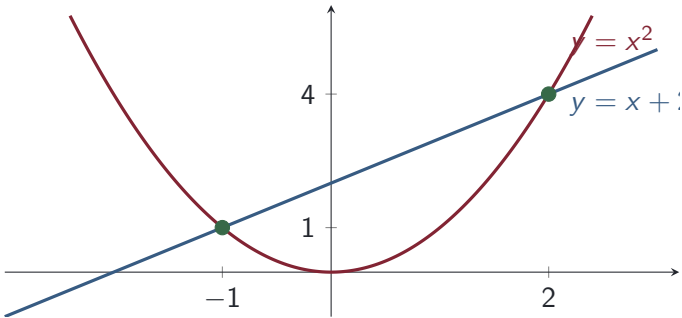
Thus

$$x = 2 \quad \text{or} \quad x = -1.$$

The intersection points are

$$\boxed{(2, 4), \qquad (-1, 1).}$$

The same example: see the intersections



Algebra gives the coordinates; the graph explains what the equation $f(x) = g(x)$ means.

The sign of a function

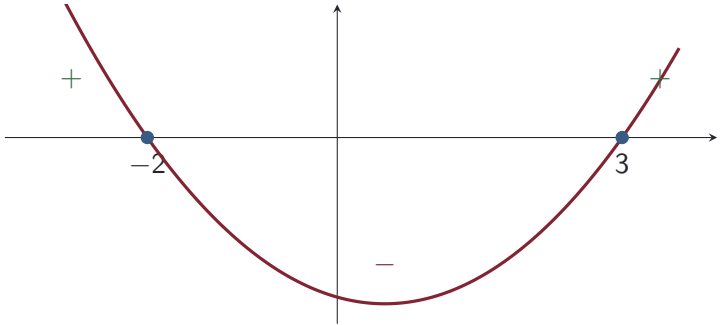
For $x \in D_f$:

$$f(x) > 0 \iff \text{the graph lies above the } x\text{-axis},$$
$$f(x) = 0 \iff \text{the graph lies on the } x\text{-axis},$$
$$f(x) < 0 \iff \text{the graph lies below the } x\text{-axis}.$$

Sign study

Determine the subsets of the domain on which f is positive, negative or zero.

Reading sign directly from a graph



For $f(x) = (x+2)(x-3)$:

$$f > 0 \text{ on } (-\infty, -2) \cup (3, +\infty), \qquad f < 0 \text{ on } (-2, 3).$$

A systematic sign-chart method

For a factorized expression:

1. Determine the domain.
2. Find the zeros of each factor.
3. Add any excluded points from the domain.
4. Put all critical values on the real line in increasing order.
5. Determine the sign of each factor on each interval.
6. Multiply or divide the signs.
7. Add the zeros when the requested inequality is non-strict.

Critical values

Zeros and excluded values divide the real line into intervals on which the sign cannot change unless a factor changes sign.

Linear factors: the basic building block

For

$$f(x) = x - a,$$

we have

x	$(-\infty, a)$	a	$(a, +\infty)$
$x - a$	$-$	0	$+$

For

$$f(x) = a - x = -(x - a),$$

the signs are reversed.

Why factorization is useful

Once a polynomial or rational function is written as a product of simple factors, its sign is obtained by combining these elementary patterns.

Polynomial sign: a complete example

Study the sign of

$$f(x) = x^2 - x - 6 = (x+2)(x-3).$$

Critical values: -2 and 3.

x	$(-\infty,-2)$	-2	$(-2,3)$	3	$(3,+\infty)$
$x+2$	$-$	0	$+$	$+$	$+$
$x-3$	$-$	$-$	$-$	0	$+$
$f(x)$	$+$	0	$-$	0	$+$

Hence

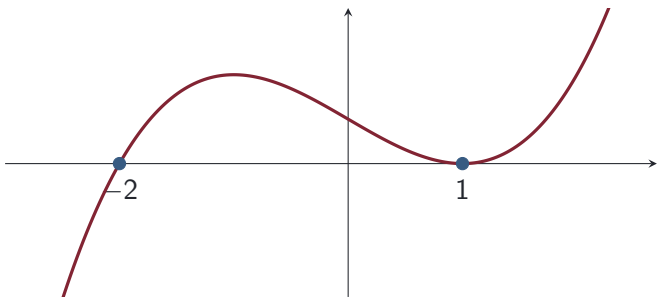
$$\boxed{f(x) \geq 0 \iff x \in (-\infty,-2] \cup [3,+\infty).}$$

Even multiplicity: the sign may not change

Consider

$$f(x) = (x-1)^2(x+2).$$

The zero $x = 1$ has multiplicity 2.



At $x = 1$ the graph touches the axis but does not cross it. The sign does **not** change.

Rational functions: zeros and forbidden points

For

$$f(x) = \frac{N(x)}{D(x)},$$

we must distinguish:

- zeros: $N(x) = 0$ with $D(x) \neq 0$;
- excluded values: $D(x) = 0$;
- sign: combine numerator and denominator signs.

Important

An excluded point is never a zero of the function, even if a factor cancels algebraically after simplification.

Rational sign: first example

Study

$$f(x) = \frac{x-2}{x+1}.$$

Domain:

$$D_f = \mathbb{R} \setminus \{-1\}.$$

Zero:

$$x = 2.$$

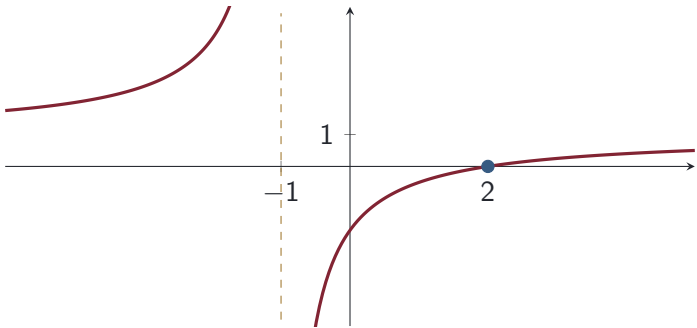
Sign chart:

x	$(-\infty,-1)$	-1	$(-1,2)$	2	$(2,+\infty)$
$x-2$	$-$	$-$	$-$	0	$+$
$x+1$	$-$	0	$+$	$+$	$+$
$f(x)$	$+$	$\times$	$-$	0	$+$

Thus

$$f(x) > 0 \iff x \in (-\infty,-1) \cup (2,+\infty).$$

Rational sign: connect table and graph



The excluded point $x = -1$ separates two branches; the zero $x = 2$ is where the graph crosses the x-axis.

Rational sign: a richer example

Study the sign of

$$f(x) = \frac{(x-1)(x+2)}{x-3}.$$

Critical values:

$$-2, \qquad 1, \qquad 3 \text{ (excluded)}.$$

x	$(-\infty,-2)$	-2	$(-2,1)$	1	$(1,3)$	3	$(3,+\infty)$
$x+2$	$-$	0	$+$	$+$	$+$	$+$	$+$
$x-1$	$-$	$-$	$-$	0	$+$	$+$	$+$
$x-3$	$-$	$-$	$-$	$-$	$-$	0	$+$
f	$-$	0	$+$	0	$-$	$\times$	$+$

Radical functions and sign

A square root is never negative:

$$\sqrt{g(x)} \geq 0$$

whenever it is defined.

- $\sqrt{g(x)} = 0$ exactly when $g(x) = 0$.
- $\sqrt{g(x)} > 0$ exactly when $g(x) > 0$.

Example:

$$f(x) = \sqrt{x-2}.$$

Then

$$D_f = [2, +\infty), \qquad f(2) = 0, \qquad f(x) > 0 \text{ for } x > 2.$$

The sign of a logarithm

For $u > 0$:

$$\ln u < 0 \iff 0 < u < 1,$$

$$\ln u = 0 \iff u = 1,$$

$$\ln u > 0 \iff u > 1.$$

This is different from the domain condition.

The domain requires only $u > 0$. To study the sign of $\ln u$, compare the argument with 1.

Logarithmic sign: example

Let

$$f(x) = \ln(x-1).$$

Domain:

$$x - 1 > 0 \quad \Longrightarrow \quad x > 1.$$

Zero:

$$\ln(x-1) = 0 \iff x - 1 = 1 \iff x = 2.$$

Sign:

$$f(x) < 0 \text{ for } 1 < x < 2,$$

$$f(x) > 0 \text{ for } x > 2.$$

Exponential and absolute-value functions

For every real x,

$$e^x > 0.$$

More generally, if $a > 0$,

$$a^x > 0.$$

Thus exponential functions have no zeros. For every admissible x,

$$|g(x)| \geq 0.$$

Moreover,

$$|g(x)| = 0 \iff g(x) = 0.$$

These structural facts often make sign analysis immediate.

Combined example: domain, zeros and sign

Consider

$$f(x) = \frac{(x-2)\sqrt{x+1}}{x-3}.$$

First the domain:

$$x + 1 \geq 0, \qquad x \neq 3,$$

so

$$D_f = [-1, 3) \cup (3, +\infty).$$

Zeros:

$$x = -1 \quad \text{and} \quad x = 2.$$

Since $\sqrt{x+1} > 0$ for $x > -1$, the sign away from $x = -1$ is controlled by

$$\frac{x-2}{x-3}.$$

Combined example: sign chart

For

$$f(x) = \frac{(x-2)\sqrt{x+1}}{x-3},$$

we obtain

x	-1	$(-1,2)$	2	$(2,3)$	3	$(3,+\infty)$
$x-2$	$-$	$-$	0	$+$	$+$	$+$
$\sqrt{x+1}$	0	$+$	$+$	$+$	$+$	$+$
$x-3$	$-$	$-$	$-$	$-$	0	$+$
f	0	$+$	0	$-$	$\times$	$+$

Thus

$$f(x) \geq 0 \iff x \in [-1,2] \cup (3,+\infty).$$

A useful workflow for the first part of function study

Given $f(x)$:

1. Find the **domain**.
2. Check symmetry if useful.
3. Find the y-**intercept**: evaluate $f(0)$ if possible.
4. Find the x-**intercepts**: solve $f(x) = 0$.
5. Factor the expression when possible.
6. Build a **sign chart** using zeros and excluded points.
7. Compare the algebraic result with the expected graph.

Common mistakes

- Solving $f(x) = 0$ before checking the domain.
- Calling a denominator zero an x-intercept.
- Forgetting that a cancelled factor may still correspond to an excluded input.
- Assuming that the sign always changes at a zero.
- Including endpoints in a strict inequality such as $f(x) > 0$.
- Treating $\ln g(x) > 0$ as merely $g(x) > 0$ instead of $g(x) > 1$.
- Drawing a sign chart without ordering the critical values first.

Guided exercise A

Exercise For

$$f(x) = \frac{x^2 - 4}{x - 1},$$

find:

1. the domain;
2. the intersections with the coordinate axes;
3. the intervals where $f(x) > 0$ and $f(x) < 0$.

Guided exercise A: solution

Solution

$$D_f = \mathbb{R} \setminus \{1\}, \qquad x^2 - 4 = (x-2)(x+2).$$

x-intercepts:

$$(-2, 0), \qquad (2, 0).$$

y-intercept:

$$f(0) = 4 \quad \Longrightarrow \quad (0, 4).$$

Critical values: $-2, 1, 2$.

$$f > 0 \text{ on } (-2, 1) \cup (2, +\infty),$$

$$f < 0 \text{ on } (-\infty, -2) \cup (1, 2).$$

Guided exercise B

Exercise Study domain, zero and sign of

$$g(x) = \ln\left(\frac{x+2}{x-1}\right).$$

Separate two questions

First require the logarithm argument to be positive. Then, inside the domain, compare the argument with 1 to determine the sign of the logarithm.

Guided exercise B: domain

We first require

$$\frac{x+2}{x-1} > 0.$$

Critical values are -2 and 1. The quotient is positive on

$$(-\infty, -2) \cup (1, +\infty).$$

Thus

$$\boxed{D_g = (-\infty, -2) \cup (1, +\infty).}$$

Now the sign of g must be studied only inside this set.

Guided exercise B: zero and sign

Zero:

$$\ln\left(\frac{x+2}{x-1}\right) = 0 \iff \frac{x+2}{x-1} = 1.$$

But this equation gives

$$x + 2 = x - 1,$$

which is impossible. Therefore g has no zeros. Moreover,

$$\frac{x+2}{x-1} = 1 + \frac{3}{x-1}.$$

Hence it is < 1 for $x < -2$ and > 1 for $x > 1$. Therefore

$$g(x) < 0 \text{ on } (-\infty, -2), \qquad g(x) > 0 \text{ on } (1, +\infty).$$

Independent practice

Determine domain, zeros and sign.

1. $f(x) = \dfrac{x+1}{x-4}$.
2. $g(x) = x(x-2)^2(x+3)$.
3. $h(x) = \sqrt{x^2-9}$.
4. $p(x) = \ln(4-x)$.
5. $q(x) = \dfrac{\sqrt{x-1}}{x+2}$.
6. Find the intersections of $y = x^2 - 1$ and $y = 2x + 2$.

Independent practice: answers

1. $D = \mathbb{R} \setminus \{4\}$; zero -1; positive on $(-\infty, -1) \cup (4, +\infty)$.
2. Zeros $-3, 0, 2$; the sign does not change at the double zero 2.
3. $D = (-\infty, -3] \cup [3, +\infty)$; zeros ± 3; positive elsewhere in the domain.
4. $D = (-\infty, 4)$; zero 3; positive for $x < 3$, negative for $3 < x < 4$.
5. $D = [1, +\infty)$; the denominator never vanishes there; zero 1; positive for $x > 1$.
6. $x^2 - 1 = 2x + 2 \Rightarrow x = 3, -1$; points $(3, 8)$ and $(-1, 0)$.

What we have built so far

For a one-variable function we can now investigate:

$$\boxed{\text{domain}} \longrightarrow \boxed{\text{intercepts and zeros}} \longrightarrow \boxed{\text{sign}}.$$

These are the first layers of a complete function study.

Next step: limits

The graph also has a **local behaviour near a point** and a **long-run behaviour for large** $|x|$. Limits give the mathematical language for both.